Algebra 1 — Name___<u>Answer key</u>______
Unit 9 — Date____________________
Lesson 3 Homework — Period___________________

Use the information provided in each table to write the equations in vertex form and standard form.

1. Identify the vertex.

x	-2	-1	0	1	2
y	31	16	7	4	7

(__1__, __4__)

2. Write the equation for this quadratic in vertex form and standard form.

a. Vertex form:

$(1,4)$
$(0,7)$

$7 = a(0-1)^2 + 4$
$7 = a + 4$
$3 = a$

$\boxed{y = 3(x-1)^2 + 4}$

b. Standard form:

$y = 3(x-1)(x-1) + 4$
$(3x-3)(x-1) + 4$
$3x^2 - 3x - 3x + 3 + 4$
$\boxed{y = 3x^2 - 6x + 7}$

3. Identify the vertex.

(__-2__, __-8__)

x	y
-4	-6
-3	-7.5
-2	-8
-1	-7.5
0	-6

4. Write the equation for this quadratic in vertex form and standard form.

a. Vertex form:

$(-2,-8)$
$(-4,-6)$

$-6 = a(-4+2)^2 - 8$
$-6 = 4a - 8$
$\frac{2}{4} = \frac{\cancel{4}a}{\cancel{4}} \rightarrow a = \frac{1}{2}$

$\boxed{y = \frac{1}{2}(x+2)^2 - 8}$

b. Standard form:

$y = \frac{1}{2}(x+2)(x+2) - 8$
$\left(\frac{1}{2}x + 1\right)(x+2) - 8$
$\frac{1}{2}x^2 + 1x + 1x + 2 - 8$
$\boxed{y = \frac{1}{2}x^2 + 2x - 6}$

5. Identify the vertex.

(_2_, _0_)

x	y
0	-8
1	-2
2	0
3	-2
4	-8
5	-18

6. Write the equation for this quadratic.

a. Vertex form:

$(2,0)$ $\quad -8 = a\,(0-2)^2 + 0$

$(0,-8)$ $\quad -8 = 4a$

$-2 = a$

$\boxed{y = -2\,(x-2)^2}$

b. Standard form:

$y = -2\,(x-2)(x-2)$

$= (-2x+4)(x-2)$

$-2x^2 + 4x + 4x - 8$

$\boxed{y = -2x^2 + 8x - 8}$

7. Identify the vertex.

(_4_, _1_)

x	1	2	3	4	5
y	$\frac{13}{4}$	2	$\frac{5}{4}$	1	$\frac{5}{4}$

8. Write the equation for this quadratic.

a. Vertex form:

$(4,1)$ $\quad 2 = a\,(2-4)^2 + 1$

$(2,2)$ $\quad 2 = 4a + 1$

$1 = 4a \rightarrow \frac{1}{4} = a$

$\boxed{y = \frac{1}{4}\,(x-4)^2 + 1}$

b. Standard form:

$y = \frac{1}{4}\,(x-4)(x-4) + 1$

$= \left(\frac{1}{4}x - 1\right)(x-4) + 1$

$= \frac{1}{4}x^2 - 1x - 1x + 4 + 1$

$\boxed{y = \frac{1}{4}x^2 - 2x + 5}$

9. Identify the vertex.

(_0_, _3_)

10. Write the equation for this quadratic.

x	y
-1	0
0	3
1	0
2	-9
3	-24

a. Vertex form:

$(0,3)$ $\quad 0 = a\,(1-0)^2 + 3$

$(1,0)$ $\quad 0 = a + 3$

$-3 = a$

$\boxed{y = -3(x)^2 + 3}$

b. Standard form:

$y = -3(x)(x) + 3$

$= (-3x)(x) + 3$

$\boxed{y = -3x^2 + 3}$